

# ARNAV GUPTA

Bandipur, Siraha, Nepal

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## About

Independent ML researcher whose work is animated by a structural question: when does the spectral and geometric character of a learned representation determine what a learning system can and cannot do—and what does that imply about when pipelines will fail? Two preprints pursue this from complementary angles. The first (arXiv, 2026) introduces a label-free saturation index  $S(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K$ , grounded in classical high-dimensional covariance estimation, that predicts annotation diminishing returns in binary few-shot classification without test labels or a trained classifier. The second (Research Square, 2026) provides the first controlled empirical documentation of verifier exploitation in NLI-guided iterative refinement—a failure mode in which a system learns to game its own verifier rather than improve factual faithfulness. Both conducted without institutional affiliation, advisor, or supervised compute, from Nepal, at fifteen.

## Publications & Preprints

### A Spectral Phase Diagram for Binary Few-Shot Classification: Intrinsic Dimensionality, Geometric Saturation, and Representational Diagnosis

- Arnav Gupta. Independent Researcher, Nepal. arXiv preprint, 2026. [doi.org/10.48550/arXiv.2606.24903](https://doi.org/10.48550/arXiv.2606.24903)
- Introduced the **saturation index**  $S(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K$ —a closed-form, label-free scalar grounded in classical high-dimensional covariance estimation theory—that predicts when labeling additional support examples yields diminishing returns in binary few-shot classification, computable in  $O(d^3)$  time with no test labels and no trained classifier.
- Established a three-phase diagram across  $N=246$  doubling-pair observations from seventeen binary tasks and six datasets; within-task Spearman median  $\rho=0.811$  (IQR [0.720, 0.896]) across sixteen of seventeen tasks, pooled  $\rho=0.548$  ( $p=1.1 \times 10^{-20}$ ); Algorithm 1 achieves AUC=0.752 as a binary stopping rule.
- Proved (Proposition 1, Corollary 1.1) that the index falls below a threshold precisely when the sample covariance is well-concentrated around the population covariance under sub-Gaussian class-conditionals; demonstrated that asymptotic effective rank and peak accuracy are geometrically orthogonal ( $r_s=0.380$ ,  $p=0.133$ ,  $N=17$ ), separating annotation cost from accuracy ceiling as independent design concerns.

### Verifier Exploitation in NLI-Guided Iterative Refinement: A Controlled Empirical Analysis

- Arnav Gupta. Independent Researcher, Nepal. Research Square preprint, 2026. [doi.org/10.21203/rs.3.rs-10187492/v1](https://doi.org/10.21203/rs.3.rs-10187492/v1)
- First controlled empirical documentation of verifier exploitation in zero-gradient, prompt-only iterative refinement: a second revision cycle inflated SummaCConv by +0.185 while collapsing BARTScore<sub>s→d</sub> by −2.566 nats, confirmed by Wilcoxon  $W=0$  ( $p=2.68 \times 10^{-83}$ ) across all 498 instances without exception.
- Identified truncation-exploiting content removal as the exploitation mechanism; formalized a three-condition, annotation-free detection protocol applicable to any NLI-guided iterative refinement pipeline.
- AnchorSum achieves a 6.3% relative SummaCConv inconsistency reduction over 498 Multi-News instances ( $p=4.49 \times 10^{-28}$ ); outperforms fine-tuned baselines (BART, PEGASUS, PRIMERA) across all LLM-as-judge dimensions in a dual-judge jury evaluation.

## Education

### Sagarmatha Secondary School

Grade 10 | GPA: 3.87 / 4.00 (Outstanding)

Mirchaiya, Siraha, Nepal

Expected SEE: 2026

## Experience

### Suvidha Foundation

Machine Learning Intern

September 2025 – November 2025

Remote

- Researched multi-document abstractive summarization with transformer-based models under mentorship of a PhD in NLP.
- Built a proof-of-concept summarization system incorporating hierarchical context aggregation across multiple source documents.
- Participated in community outreach programs supporting underserved populations alongside technical work.

## Programs & Fellowships

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### Harvard Computer Society AI Bootcamp

2025

*Artificial Intelligence Fellow*

*Remote*

- Completed an intensive AI fellowship collaborating with Harvard students, focused on frontier AI systems, modern architectures, and real-world deployment challenges.
- Studied Transformer architecture in depth and implemented one from scratch; covered embeddings, positional encoding, self-attention, cross-attention, and position-wise feed-forward networks.
- Explored Reinforcement Learning foundations: MDPs, Value Iteration, Policy Iteration, SARSA, Q-Learning, and Multi-Armed Bandits.
- Analyzed frontier LLM architectures (ChatGPT, Claude, Gemini, Perplexity) with focus on hallucination mitigation strategies and inference-time reliability.
- Completed hands-on projects: fine-tuned BERT, built a mini-LLM, implemented Vision Transformers, and constructed neural networks from first principles.

## Projects

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### NietzscheGPT | *Python, PyTorch, Hugging Face Transformers* | [Hugging Face](#)

2025

- Fine-tuned GPT-2 Medium (355M parameters) on the complete English works of Nietzsche (25 books, ~2.2M tokens) to generate philosophical text in his style; trained on Apple M3 Pro via MPS backend in ~4 hours and published on Hugging Face.

### Transformer from Scratch | *Python, PyTorch, NumPy* | [GitHub](#)

2025

- Implemented the full Transformer encoder from first principles—sinusoidal positional encoding, multi-head self-attention, position-wise FFN, residual connections, layer normalization—trained on a synthetic autoregressive task; structured as a modular notebook bridging mathematical derivations and working code.

### 10D Hypercube Visualization | *Python, NumPy, Scikit-learn, Matplotlib* | [GitHub](#)

2026

- Constructed and visualized a 10-dimensional hypercube (1,024 vertices; 5,120 edges) using PCA, t-SNE, and layer-stratified views; generated comparative panels illustrating how geometric structure is preserved or distorted under each projection.

### House Price Prediction | *Python, Scikit-learn, XGBoost, Pandas* | [GitHub](#)

2025

- Regression pipeline on Ames Housing (Kaggle): EDA, feature engineering (composite features, log transforms, ordinal encoding); tuned XGBoost (CV  $\text{RMSE}_{\log}$  0.1229) outperforming Ridge baseline (0.1445) via GridSearchCV.

### Live Emotion Detection | *Python, DeepFace, OpenCV, TensorFlow, MTCNN* | [GitHub](#)

2025

- Real-time 7-class facial emotion classifier using DeepFace with MTCNN detection; temporal smoothing via 10-frame sliding window, frame-skipping for throughput, overlaid confidence bars and FPS counter.

## Technical Skills

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**Languages:** Python, C++, SQL

**Libraries/Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, Matplotlib, XGBoost, OpenCV, DeepFace

**Tools & Platforms:** Git/GitHub, Docker, Jupyter, VS Code, CUDA, Firebase, MySQL, REST APIs, Zotero, Obsidian

**Typesetting:** L<sup>A</sup>T<sub>E</sub>X, Markdown

## Mathematics Coursework (MIT OpenCourseWare)

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- **18.01 – Single Variable Calculus** (Completed): Differential and integral calculus with rigorous problem-solving.
- **18.06 – Linear Algebra** (In Progress): Matrix theory, eigendecomposition, orthogonality, and applications to machine learning.